Accuracy and Imbalance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Accuracy and Imbalance

> Prepare Accuracy and Imbalance: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain why overall **accuracy** can hide failure on a less common class

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/12, ai-4/16. Reuse a familiar case for **recall** or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and **prediction** pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix [[2,1],[2,3]].

Train a Classifier

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Connect **training**-only scaling and neighbour **prediction** in one pipeline

Reference example, not the only acceptable wording: fit learns **training** minima and ranges and stores scaled **training** rows with IDs and labels. predict takes a raw feature pair and the fitted state.

Evaluation compares a **prediction** with its reference label afterward. Validation references belong only in evaluation, not in **fitting** or the neighbour pool.

Baselines First

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a majority label baseline using **training** labels only

Reference example, not the only acceptable wording: **Training** has four clear and two blocked, so the constant is clear. A tie chooses clear under the declared classroom rule.

Validation labels score the stored **prediction**; using their majority to choose it would change the **fitting** procedure.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Model A is always healthy on 96 healthy and 4 spotted leaves. Model B finds 3 spotted but falsely flags 5 healthy. Compare **accuracy** and spotted detection.

Concept to explain

Accuracy can be dominated by a common class and hide failure on a rare but important class.

Term	Meaning
accuracy	correct predictions divided by all predictions
class imbalance	classes occur in very different amounts
minority class	less frequent target class
recall	fraction of actual positives found

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: **recall** 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. Begin with a majority guess that looks successful under overall **accuracy**, then keep both class denominators visible.

Introduce **balanced accuracy** as an equal average of two correct fractions, without treating it as a universal replacement for task-specific evaluation.

Readiness answer

They are in the blocked-reference row and clear-**prediction** column: the lower-left cell. These are missed blocked cases.

Printed lesson and workbook

Twenty counters with sixteen labelled clear and four blocked

Blank confusion matrices and fraction strips

Optional Python 3 editor with no packages required

End goal for the pictorial case

By the end, I can explain **accuracy** and imbalance using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Accuracy and Imbalance using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Compare trivial **accuracy** with minority-class **recall**.

MODEL IDEA **Accuracy** can be dominated by a common class and hide failure on a rare but important class.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Test set has 95 healthy and 5 diseased leaves. Always predicting healthy gives 95 ÷ 100 = 95% **accuracy**.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Test set has 95 healthy and 5 diseased leaves.

Step 2

STEP 2 Always predicting healthy gives 95 ÷ 100 = 95% **accuracy**.

Step 3

STEP 3 It finds 0 of 5 diseased leaves, so diseased **recall** = 0%.

Step 4

STEP 4 Report class counts and **recall**; redesign data or use only where misses are acceptable.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

A dataset has 95 healthy and 5 spotted. Model predicts healthy for all.

The teaching case

Calculate and critique **accuracy**.

Exact explanation to model

> ### Why the result follows
>
> 95/100=95%, but spotted **recall**=0/5=0%. Majority performance hides the detection failure.

Explain the added worked example

Explain the added worked example · 1

Accuracy counts all correct **prediction**s and divides by all evaluated examples. Each example has equal weight, so a common class contributes more to the score. A high result can hide failure on a rare class.

This week keeps the confusion matrix visible and checks both classes separately.

Explain the added worked example · 2

New fictional **training** data has 16 clear and 4 blocked labels. Its **majority baseline** predicts clear for every query; choose that rule from **training**, not evaluation labels.

A separate evaluation set also contains 16 clear and 4 blocked cases. A frozen candidate and the baseline are evaluated on the same 20 records.

Explain the added worked example · 3

Rows are references and columns are **prediction**s, ordered clear then blocked. Blocked is positive. Both methods get 16/20=80 percent **accuracy**. The baseline misses all four blocked cases.

The candidate finds all four but falsely flags four clear cases. The same score conceals a different tradeoff.

Explain the added worked example · 4

The **output** is baseline 80.0 50.0, then candidate 80.0 87.5. In each **output** line the first percentage is ordinary **accuracy** and the second is **balanced accuracy**.

For the candidate, the clear-class correct fraction is 12/16=75 percent and the blocked-class fraction is 4/4=100 percent. Their equal average is 87.5 percent.

Explain the added worked example · 5

For the baseline, those fractions are 16/16=100 percent and 0/4=0 percent, averaging 50 percent. The code assumes a two-by-two table of nonnegative integer counts in the stated order.

It stops when either reference class is absent. Then report the observed counts and the unavailable class result; do not invent a denominator or silently substitute zero.

Key terms

Term	Meaning
Class imbalance	Unequal numbers of examples in different classes
Majority baseline	A rule that always predicts the most common training label
Class-specific correct fraction	Correct predictions for one reference class divided by the number of examples of that class
Balanced accuracy	The average of the class-specific correct fractions with equal weight for each class

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Same overall score different class results

SUMMARY	ALWAYS CLEAR BASELINE	FROZEN CANDIDATE
Overall correct	16 of 20 is 80 percent	16 of 20 is 80 percent
Reference clear correct	16 of 16 is 100 percent	12 of 16 is 75 percent
Reference blocked correct	0 of 4 is 0 percent	4 of 4 is 100 percent

Overall accuracy weights classes by their number of evaluation examples. Balanced accuracy gives the two class-specific fractions equal weight.

Both methods score 80 percent. Explain which cases the baseline fails, how the candidate changes the errors, and why their balanced accuracies differ.

Model the reasoning aloud · 6

Balanced accuracy is a metric choice, not a repair to the classifier or a guarantee of safety. It gives classes equal influence in this summary; it does not encode the actual consequences of false alarms and missed cases.

Four blocked evaluation examples are also too few for a strong general claim.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Accuracy and Imbalance

Show how a common healthy class can hide complete failure to detect spotted leaves.

The facts we will use

Supplied facts

Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy. A candidate hasTP2, FN 0, FP 3, TN 15.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/16

Candidate decisions

	predicted spotted	predicted healthy
actual spotted	2	0
actual healthy	3	15

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/16

Accuracy and Imbalance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Score trivial baseline

Always healthy is 18/20=90% accurate but detects0/2 spotted leaves.



2. Score candidate

Candidate accuracy is 17/20=85%; spotted recall is 2/2=100%.



3. Compare consequences

Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

CHANGE ONE THING

The candidate's three false alarms are corrected without changing its detections.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Score trivial baseline	Always healthy is 18/20=90% accurate but detects0/2 spotted leaves.
2. Score candidate	Candidate accuracy is 17/20=85%; spotted recall is 2/2=100%.
3. Compare consequences	Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

Change one thing in the pictured case

Change one thing

Change: The candidate’s three false alarms are corrected without changing its detections.

Predicted result

Prediction: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

Why the result follows

Reason: False alarms move from FP to TN; neither TP nor FN changes.

Ask while showing the picture

Ask while showing the picture · 7

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Accuracy and Imbalance, reduce support deliberately and check individual reasoning before discussing answers.

Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Board sequence

1. Put the facts on the board

Write the given facts: A dataset has 95 healthy and 5 spotted. Model predicts healthy for all.

2. State the decision to explain

Write the question to explain: Calculate and critique **accuracy**.

3. Reveal the worked reasoning

Reveal after **prediction**s: 95/100=95%, but spotted **recall**=0/5=0%. Majority performance hides the detection failure.

Hold back the independent task

Keep for an independent attempt: A classifier predicts healthy for all 100 plants; 95 are healthy and 5 diseased. Compute **accuracy** and diseased **recall**.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: baseline has FP=0,FN=4, finding none of four blocked cases. Candidate has FP=4,FN=0, finding all four but adding four false alarms. Both have sixteen correct **prediction**s.

Equal averaging gives baseline (100+0)/2=50 percent and candidate (75+100)/2=87.5 percent. Exit: request the full labelled matrix, class counts, sampling method, **training**-derived baseline and task consequences.

The worked always-clear rule misses all four blocked cases.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve false negatives and reference-row totals from Week 13.
7 to 18 minutes	Use counters to score the always-clear baseline and complete W1.

Stage	Teacher action
18 to 32 minutes	Compare the candidate matrix. Calculate ordinary and class-specific correct fractions in W2.
32 to 45 minutes	Trace the code and read the visual. Answer the guided question and explain equal class weighting.
Session 2 start to 8 minutes	Retrieve each denominator and why a high overall score can hide rare-class failure.
8 to 20 minutes	Complete W3 with a changed class mix. Separate the arithmetic assumption from claims about deployment.
20 to 35 minutes	Collect W4 independently before support. Require fractions before rounded percentages and the error tradeoff.
35 to 45 minutes	Use W5 and the exit question to discuss missing classes. Offer the fresh retry after feedback and record support.

Reduce help deliberately

Completion practice

Completion task: For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

Reasoning to listen for

Teacher check: **Accuracy** 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: A common class can dominate the overall count. Inspect each reference-class denominator and its errors.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

It misses every spotted leaf. **Accuracy** is dominated by18 healthy cases; report class-specific errors and the intended-use consequences.

Changed-case check: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set. False alarms move from FP to TN; neither TP nor FN changes.

Changed-case reasoning: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set. False alarms move from FP to TN; neither TP nor FN changes.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Accuracy and Imbalance investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Imbalance Simulation: compare majority **prediction** with a model using **accuracy**, **recall**, and the actual error costs.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A classifier with 95% **accuracy** is excellent in every practical situation.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

High **accuracy** means both classes work well: A common class can dominate the overall count. Inspect each reference-class denominator and its errors.

Balanced accuracy fixes the model: It changes the summary weighting, not any **prediction** or underlying error.

An absent class has zero measured performance: Its fraction has a zero denominator and is unavailable. Report missing coverage rather than inventing a value.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Accuracy and Imbalance questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

> **Teacher key · Worked case answer**

95/100=95%, but spotted **recall**=0/5=0%. Majority performance hides the detection failure.

Independent question

Independent question · withhold the key

A classifier predicts healthy for all 100 plants; 95 are healthy and 5 diseased. Compute **accuracy** and diseased **recall**.

Expected independent answer

Teacher key · Expected independent answer

Accuracy is 95/100 = 95%. Diseased **recall** is 0/5 = 0%; the model misses every diseased plant.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “High overall **accuracy** rules out a serious minority-class failure.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

A **accuracy**=96%, **recall**=0%. B has TP=3,FN=1,FP=5,TN=91; **accuracy**=94%, **recall**=75%. Higher **accuracy** alone favours A despite missing all positives. The preferred tradeoff depends on task consequences and review capacity.

Explanations for the additional workbook examples

W1 Score the majority baseline

Workbook question · W1

Training has 16 clear and 4 blocked labels. State the majority rule. On the worked evaluation, give its matrix, **accuracy**, false positives and false negatives with blocked positive. Explain why 80 percent is incomplete.

Teacher reasoning · W1

The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4. It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

W2 Compare both class fractions

Workbook question · W2

For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives. Compare with the baseline.

Teacher reasoning · W2

Accuracy 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

W3 Explain a changed class mix

Workbook question · W3

For the illustrative 8-clear 8-blocked set, use baseline [[8,0],[8,0]] and candidate [[6,2],[0,8]]. Calculate ordinary and **balanced accuracy** for each. Explain what stayed fixed and what this example does not prove.

Teacher reasoning · W3

Baseline ordinary 8/16=50 percent and balanced (100+0)/2=50 percent. Candidate ordinary 14/16=87.5 percent and balanced (75+100)/2=87.5 percent. Within-class fractions stayed fixed while class counts changed.

This arithmetic assumption does not prove unchanged real-world behavior on a new population.

W4 Assess a new comparison independently

Workbook question · W4

New **training** labels have a clear majority. A separate evaluation has 18 clear and 2 blocked records.

The always-clear baseline is [[18,0],[2,0]] and candidate is [[15,3],[1,1]], with reference rows and clear then blocked on both axes. Calculate ordinary **accuracy**, both class fractions and **balanced accuracy** for each.

Give candidate FP and FN, then explain why the metrics do not choose an automatic winner.

Teacher reasoning · W4

Baseline **accuracy** 18/20=90 percent; class fractions 100 percent and 0 percent; **balanced accuracy** 50 percent.

Candidate **accuracy** 16/20=80 percent; clear fraction 15/18=5/6 about 83.3 percent, blocked fraction 1/2=50 percent; **balanced accuracy** (5/6+1/2)/2=2/3 about 66.7 percent. Candidate FP=3,FN=1.

It finds one blocked case but adds three false alarms; task consequences and the small sample matter.

W5 Limit claims from missing classes

Workbook question · W5

An evaluation contains ten clear cases and no blocked cases. All **prediction**s are clear. State ordinary **accuracy**, explain why blocked performance is unavailable, and describe how our code handles the missing class.

Teacher reasoning · W5

Ordinary **accuracy** is 10/10=100 percent on the observed clear cases. The blocked-class fraction has denominator zero, so it cannot be measured.

Our two-class summary function stops with ValueError; report the missing coverage instead of claiming perfect performance on both classes.

Assess independent transfer

Independent question · withhold the key

Selected case: Validation has90 healthy and10 spotted cases. Always-healthy predicts all healthy. Another model has TP7,FN3,FP8,TN82. Compare **accuracy** and spotted **recall**.

Selected case answer

Teacher key · Selected case answer

Always-healthy **accuracy**90% but spotted **recall**0%. The other model has89% **accuracy** and**70%** spotted **recall**. Its false positives are8.

Decide using declared consequences and relevant metrics; **accuracy** alone hides missed spotted cases.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Accuracy and Imbalance to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Check Plant Classifier class balance and select metrics that keep the spotted-leaf class visible.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 17: Precision and **Recall**. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use counters and write each fraction with its reference-class total before converting to a percentage. Supply blank denominator boxes rather than answers.

Extension: express ordinary **accuracy** as the class-size-weighted average of the two correct fractions; for the candidate use (16/20)(12/16)+(4/20)(4/4)=16/20.

Require W4 baseline ordinary 90 percent and balanced 50 percent; candidate ordinary 80 percent and balanced 2/3 about 66.7 percent, with FP=3,FN=1.

Fresh retry: evaluation 12 clear and 4 blocked, baseline [[12,0],[4,0]], candidate [[9,3],[1,3]], **training** majority clear. Baseline ordinary 75 percent, class fractions 100 and 0 percent, balanced 50 percent.

Candidate ordinary 12/16=75 percent, both class fractions 75 percent, balanced 75 percent, FP=3,FN=1. Require both denominators and no automatic-winner claim.

Week 15 covers precision **recall** and denominators. Carry forward the matrix and the question answered by each fraction.

Teacher reference: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.balanced_**accuracy**_score.html (equal average of class **recall**s; reviewed 2026-09-28).

Plan the delayed return

Suggested return points in this level: ai-4/17, ai-4/19, ai-4/22. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Accuracy and Imbalance lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

> Teaching target: Explain why overall **accuracy** can hide failure on a less common class

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: For the worked candidate [[12,4],[0,4]], give ordinary **accuracy**, both **class-specific correct fraction**s, **balanced accuracy**, false positives and false negatives.

Compare with the baseline.

Supported practice answer

Reasoned practice answer: **Accuracy** 16/20=80 percent. Clear fraction 12/16=75 percent; blocked fraction 4/4=100 percent; **balanced accuracy** (75+100)/2=87.5 percent. FP=4,FN=0.

Baseline **accuracy** is also 80 percent but **balanced accuracy** is 50 percent with FP=0,FN=4.

Check every learner before discussion: Use the worked case for **Accuracy** and **class imbalance**. Proposed explanation: “High **accuracy** means both classes work well”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A common class can dominate the overall count. Inspect each reference-class denominator and its errors.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The candidate’s three false alarms are corrected without changing its detections. Require a **prediction** and a reason before comparing.

Expected result: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set. Reason: False alarms move from FP to TN; neither TP nor FN changes.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

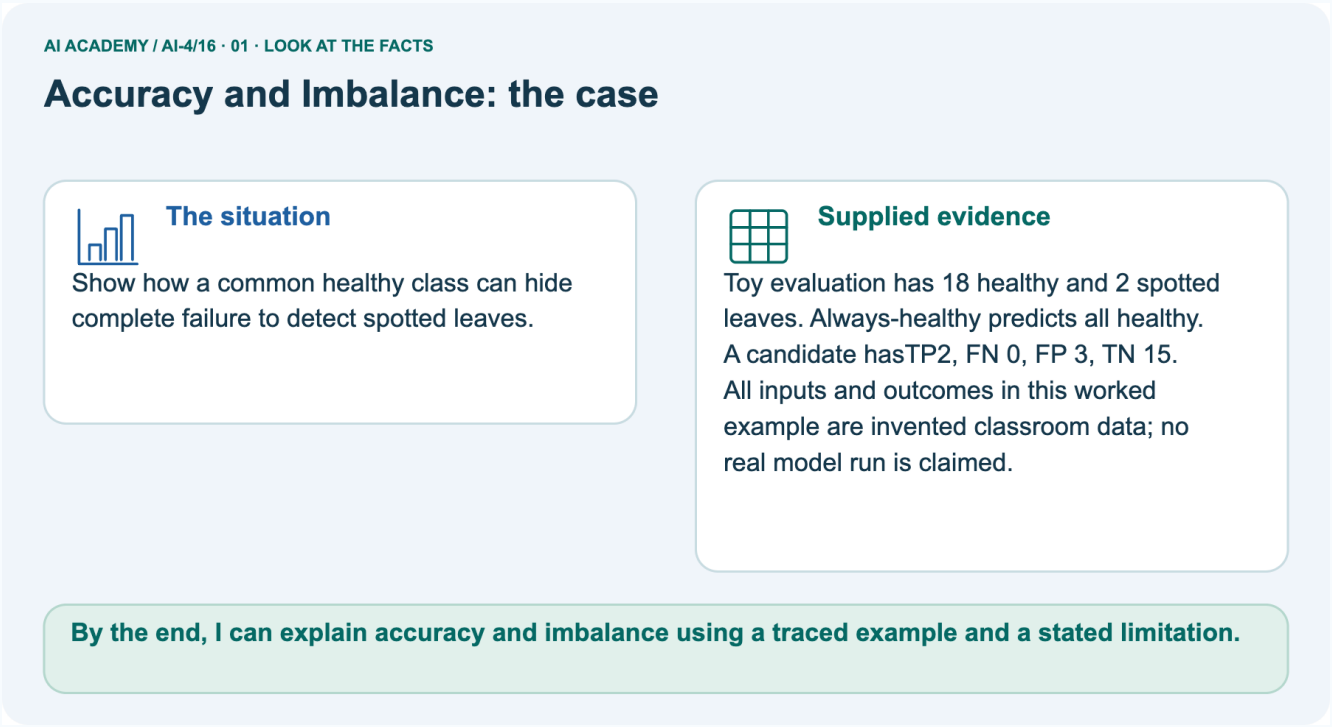
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Accuracy and Imbalance graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Accuracy and Imbalance: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Show how a common healthy class can hide complete failure to detect spotted leaves. Supplied facts: Toy evaluation has 18 healthy and 2 spotted leaves. Always-healthy predicts all healthy.

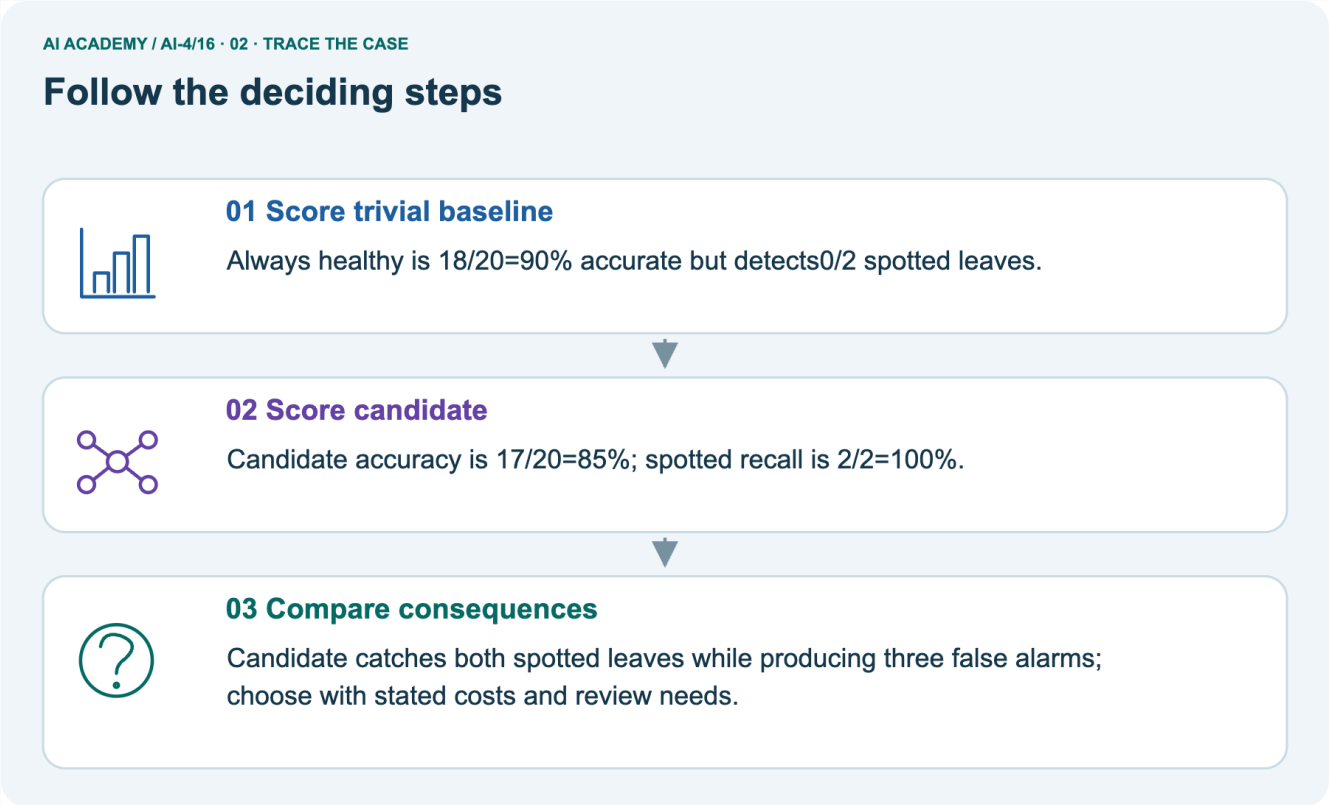
A candidate hasTP2, FN 0, FP 3, TN 15. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

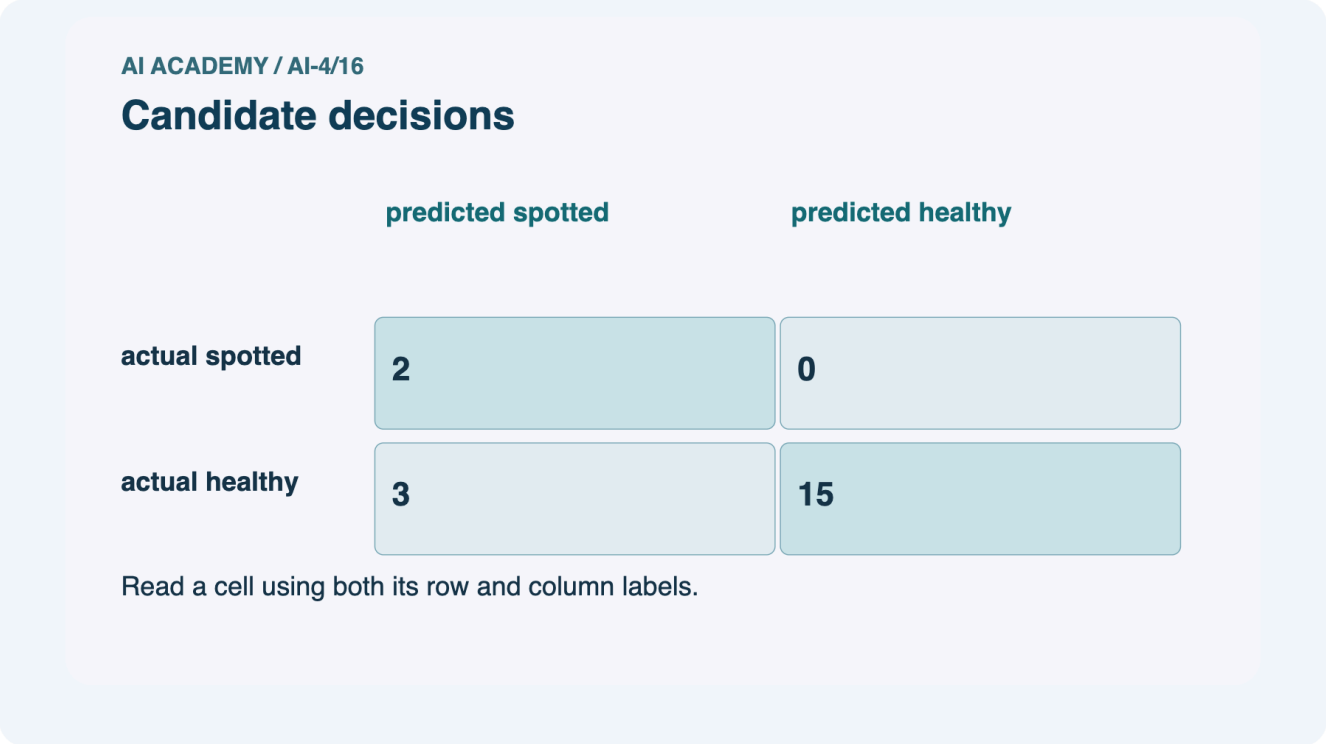
Score trivial baseline: Always healthy is 18/20=90% accurate but detects0/2 spotted leaves. → Score candidate: Candidate **accuracy** is 17/20=85%; spotted **recall** is 2/2=100%. → Compare consequences: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Candidate decisions



Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate decisions. actual spotted: predicted spotted = 2, predicted healthy = 0. actual healthy: predicted spotted = 3, predicted healthy = 15.

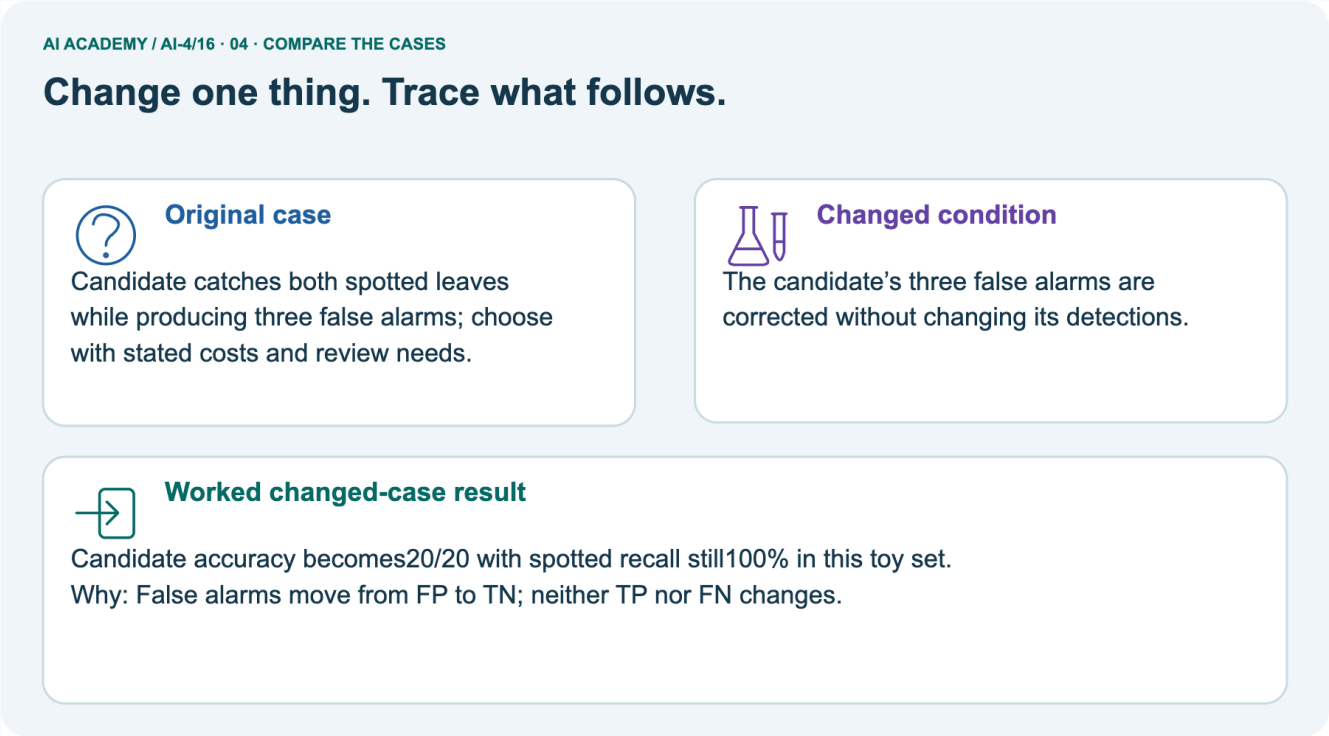
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs.

Changed condition: The candidate's three false alarms are corrected without changing its detections. Expected result: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set.

Reason: False alarms move from FP to TN; neither TP nor FN changes.

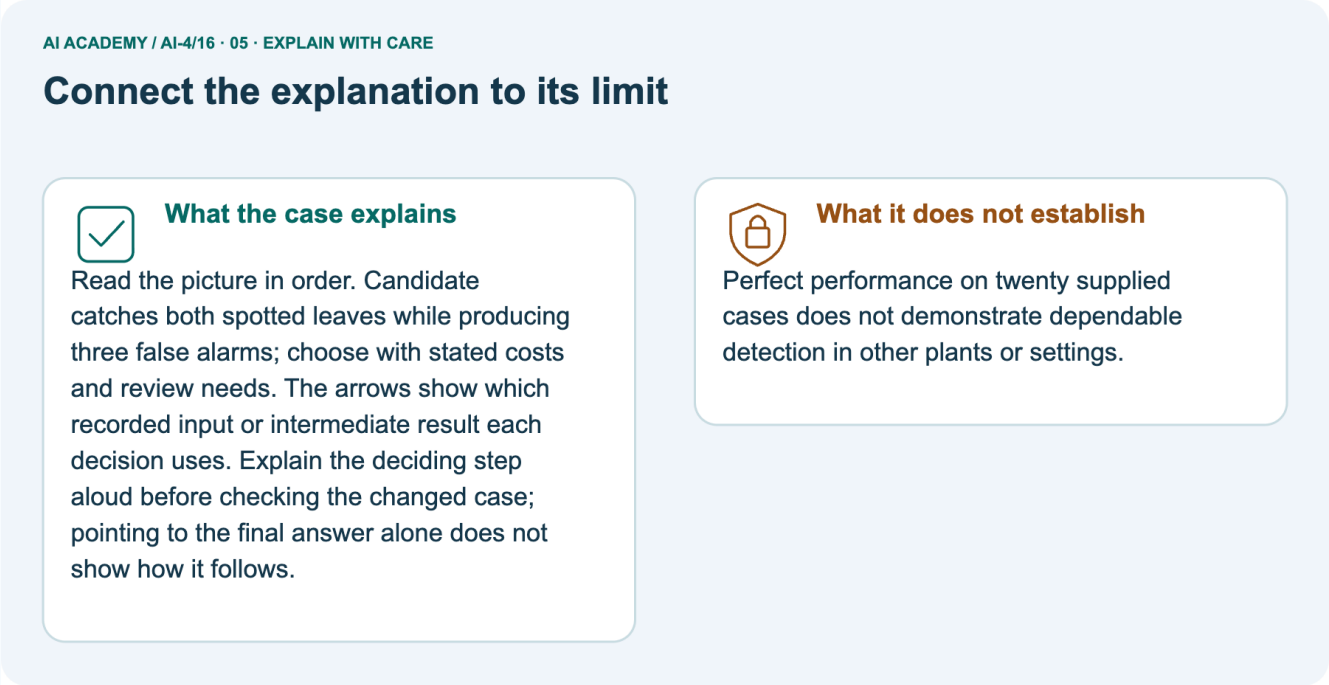
> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: False alarms move from FP to TN; neither TP nor FN changes.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Candidate catches both spotted leaves while producing three false alarms; choose with stated costs and review needs. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

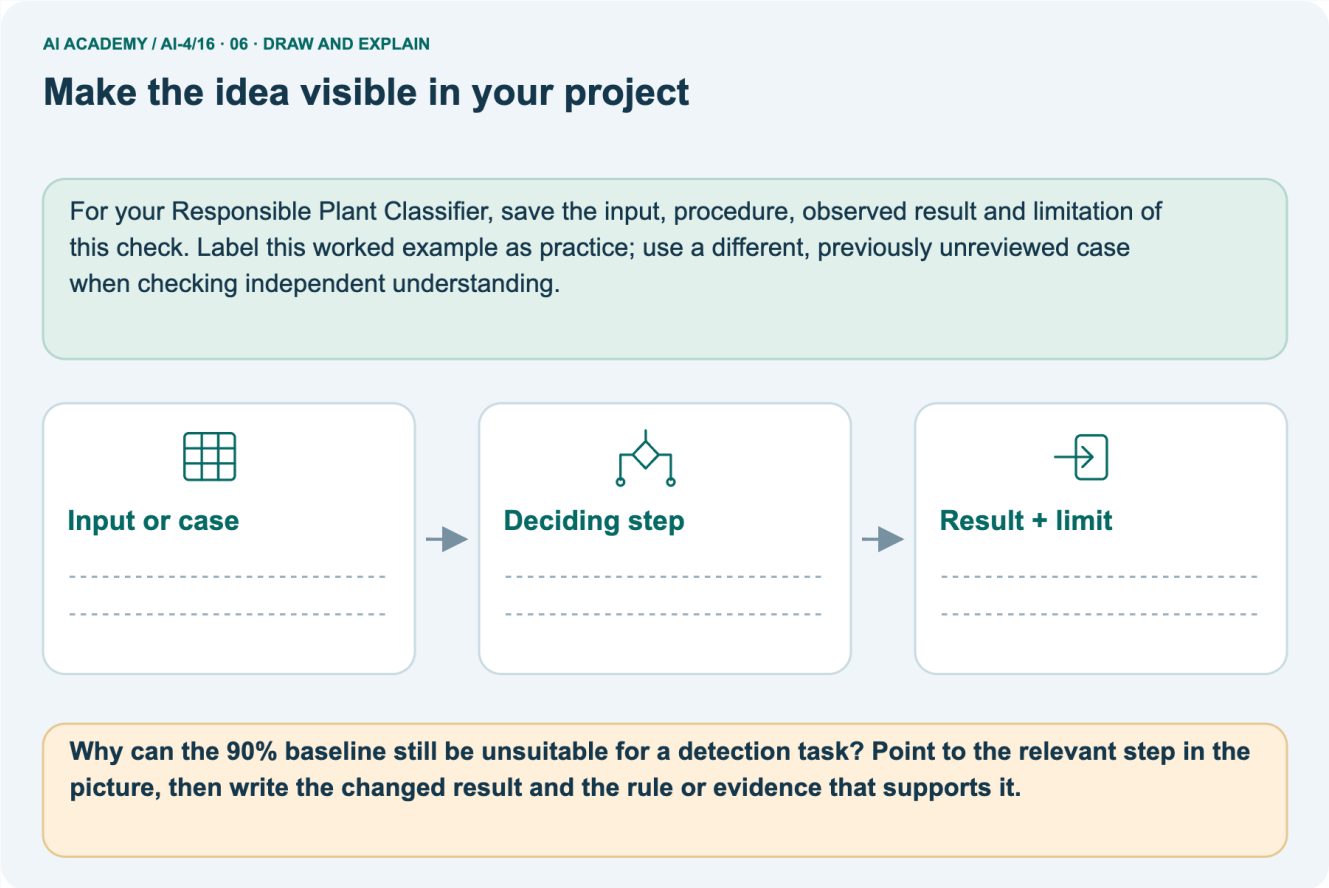
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Perfect performance on twenty supplied cases does not demonstrate dependable detection in other plants or settings.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the 90% baseline still be unsuitable for a detection task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It misses every spotted leaf. **Accuracy** is dominated by18 healthy cases; report class-specific errors and the intended-use consequences.

Changed-case check: Candidate **accuracy** becomes20/20 with spotted **recall** still100% in this toy set. False alarms move from FP to TN; neither TP nor FN changes.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.